Manual

In-Process Inspection Reports

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Contents

1 Overview – Evaluations of In-process Inspections

Purpose

This component extends the functions of the inspection requirements to include

- The display of tolerance, action and warning limits
- The visualization of the measures, errors, measured values and statistical values collected and
- the access to the document list.

Implementation Notes

The use of this component is to be recommended if the functions of the inspection requirements are to be expanded to include further detailed information.

Integration

This component predominantly serves the component "Inspection planning for in-production inspections".

Features

The following functions are available:

- Display of tolerance limits, action limits, sampling plan, etc.
- Characteristic-related display of the errors, measures, measured values and evaluations collected and the statistical values
- Visualization of the processing status using differing statuses (released, part-result, completed, etc.) and the assignment / determination of a usage decision on completion of the inspection requirements (release, special release, reject, rework, etc.)
- Display of the documents (drawings, images, videos in any format) from the document list of the characteristics, inspection plans, inspection plan characteristics, inspection requirements or inspection step characteristics

2 Inspection requirements

Overview

Menu	Quality management → In-production inspection → Inspection requirement Quality management → Goods receipt → Inspection requirement Quality management → Goods issue → Inspection requirement Quality management → Initial sample → Inspection requirement Quality management → Gage management → Inspection requirement
Transaction code	irp[1]
Function authorization	irp

Available user fields		
Where	**Object type/user field key**	**Source (type)**
Inspection requirement: Table and detail view	Production: CPAN/FEP Goods receipt: CPAN/WEP Goods issue: CPAN/WAP Initial sample: CPAN/EMU Gage management: CPAN/PMV	QM
Inspection step characteristics: Table and detail view	Production: CPAUMM/FEP Goods receipt: CPAUMM/WEP Goods issue: CPAUMM/WAP Initial sample: CPAUMM/EMU Gage management: CPAUMM/PMV	QM
Inspection points: Table and detail view	CPANUMP/PPUNKT	QM
How to configure user fields?	Which user field types are available?	

Purpose

The inspection requirement in the inspection process corresponds to the production order in BDE (shop floor data collection). The inspection requirement includes one or several inspection steps (e.g. one inspection step per operation). When an inspection requirement is generated, the system uses the respective inspection plan to create the inspection steps. The inspection requirement includes the general information, such as order or article number. One level below the inspection requirement, there is the inspection step. The inspection step then specifies the characteristics to be checked. To collect measured values, you therefore select the inspection step or the operation of the production order (that includes the inspection step) and log it on to the terminal.

As the functional requirements are almost identical in all areas (e.g. goods receipt, production, goods issue, gage management), the respective applications all have a similar structure.

[1]If you use transaction codes to call the application *Inspection requirement*, you cannot create new requirements.

If you create new inspection requirements, you are generally supported by master data catalogs. Therefore, careful maintenance of master data is very important.

Integration

You generate inspection requirements/inspection steps using inspection plans, which you have created previously. These plans are the reference the inspection is based on. When being generated, the contents of inspection plans are copied into the inspection requirements/inspection steps. The inspection plan is the reference used to create inspection requirement or inspection step. If you make changes to the inspection plan (limited options), these changes do not affect inspection requirements/inspection steps that have already been generated.

The inspection steps, in turn, are the "objects" that can be checked/inspected. In production, the inspection steps are connected to the respective operation of the production order via operation number. In gage management, the inspection step actually is a calibration order.

Almost all reports and evaluations reference the data of the inspection requirements/inspection steps and the included inspection step characteristics. To restrict the data in reports, you can filter by the article number, order number, operation, and inspection step characteristic, for example.

In gage management, the inspection requirement is automatically generated from the activity calendar. This generation is triggered by a user action. In exceptional cases, inspection requirements can also be generated directly in the “inspection requirement” application. Here, you must manually enter the inspection plan number.

Requirements

Before you can create inspection requirements/inspection steps, you must first create inspection plans for the required area/context. You also require some master data. It depends on the planned use which master data you must maintain.

Selection criteria

The following list shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

Selection list of the configured areas of in-production inspection, goods receipt or goods issue. By default, the following areas are available:
In-production inspection: Production
Goods receipt: Goods receipt
Goods issue: Goods issue
Initial sample: Initial sample inspection
Gage management: Gage management

Inspection requirement no.

The inspection requirement number uniquely identifies the inspection requirement. HYDRA automatically generates the inspection requirement number, which is unique.

Status

Open a status list to filter by the different inspection requirement statuses. This way, you can limit the list of inspection requirements to the completed inspection requirements, for example.

Skip lot (available for the goods receipt and goods issue only)

Due to dynamic modifications, an inspection requirement can be classified as "skip lot" and, as a result, no inspection is performed.

Order

Number of the respective production order or calibration order.

Article number

You can directly enter the number or open the article catalog where you can select an entry and take over the number.

Drawing issue number

The article is uniquely identified using the combination of "article number" and "drawing issue number".

Customer number

You can directly enter the number or open the customer catalog where you can select an entry and take over the number.

Supplier number

You can directly enter the number or open the supplier catalog where you can select an entry and take over the number.

Manufacturer number

You can directly enter the number or open the manufacturer catalog where you can select an entry and take over the number.

Actual date from / to

You can enter a time limit. In production, this field is usually empty. In goods receipt, the "actual date" specifies the date when the goods are delivered.

Planned date from / to

You can enter a time limit. In production, this field is usually empty. In goods receipt, the "planned date" specifies the date when the goods are planned to be delivered (initially).

Manufacturing date from / to

You can enter a time limit.

Consider long-term data

If you enable this option, you can show archived inspection requirements of the medium-term data area.

You cannot show inspection certificats, etc. in Word format for the archived inspection requirements displayed.

Field descriptions of the *Inspection requirement*

In the following, find a description of the most important fields.

Inspection requirement

Area

The inspection requirement is created for the specified area. The application provides a selection list of the configured areas of in-production inspection, goods receipt, goods issue, initial sampling or gage management. By default, the following areas are available:

In-production inspection: Production

Goods receipt: Goods receipt

Goods issue: Goods issue

Initial sample: Initial sample inspection

Gage management: Gage management

This field is a key field. Using this field, the active inspection plan or the specified inspection plan including index is identified that is used to create the inspection requirement, the inspection steps and inspection step characteristics.

Inspection requirement no.

The inspection requirement number is automatically generated upon saving. The number is numeric and uniquely identifies an inspection requirement. It cannot be changed.

Operation, operation designation

If you use operations, you must only fill in one of the two fields "operation" or "operation designation". If you want to automatically generate an inspection requirement/an inspection order when you log on a production order, you must only fill in the field *Operation number.* Usually the operation number is the only reference required because this is the most important number in the common work plan structures.

This field can be a key field. Depending on the system configuration, this field is used to search for the active inspection plan. The generated inspection requirement, inspection steps and inspection step characteristics are then based on this plan. If the system is configured accordingly, the system generates the inspection requirement and all inspection steps at the same time for the operations that are included in the respective inspection plan.

Also in case of areas that do not have production orders (e.g. goods receipt and gage management), it might be required to specify a “fictitious” operation (e.g. 9999) depending on the system configuration. This “fictitious” operation has to be included in the respective inspection plan.

Inspection plan number / inspection plan index (specified)

You can use these fields to specify the inspection plan that is used for the generation of the inspection requirement if you do not specify key fields to search for the active inspection plan.

Date

When a new data record is created, this field includes the system date, which may be changed afterwards. Once it has been saved, this field may no longer be changed.

Article

Enter the article number. If you know the article number, you can directly enter it. If not, you can open the article catalog and identify and transfer the requested article using the filter and sort criteria. Select an article to take over the drawing issue number, the article designation, the customer article number and the drawing number from the master data record. The respective data is then displayed in the respective fields.

This field is a key field. You use this field to search for the active inspection plan that is used to generate inspection requirement, inspection steps and inspection step characteristics if no inspection plan including index has been specified before.

In gage management, this field is empty or should be left empty.

Drawing issue number

Like the article number, you can directly enter the drawing issue number.

This field is a key field. You use this field to search for the active inspection plan that is used to generate inspection requirement, inspection steps and inspection step characteristics if no inspection plan including index has been specified before.

If you directly enter the article number and the drawing issue number, this information is used upon saving to identify the master data record. Using this master data, the system can then display article name, customer article number and drawing number.

The article is uniquely identified by the combination of article and drawing issue number. It is not obligatory to define and use a drawing issue number.

If the field “Drawing issue number” is shown in gage management, you may not enter any values here.

Companies

Customer / Supplier / Manufacturer

You can directly enter the respective company number or open the master data catalog and select the respective entry. When you have selected and saved the inspection requirement, the system also enters the company name and address data.

These fields are key fields. You use these fields to search for the active inspection plan that is used to generate inspection requirement, inspection steps and inspection step characteristics if no inspection plan including index has been specified before.

Date - quantities

Actual quantity

The system uses the actual quantity to calculate the inspection scope. The calculation is based on the selected sampling scheme. This is especially true in goods receipt and goods issue if dynamic modification is enabled. In production, this field is usually empty.

Target quantity

In particular for the goods receipt, it is important to specify the target quantity in order to identify the difference between the ordered quantity and the actually delivered quantity.

Scrap

This field is only relevant if you are just using the CAQ functions of HYDRA and you are not using the shop floor data collection (BDE) module.

Rework

This field is only relevant if you are just using the CAQ functions of HYDRA and you are not using the shop floor data collection (BDE) module.

Actually produced quantity (act. prod. qty.)

This field is only relevant if you are just using the CAQ functions of HYDRA and you are not using the shop floor data collection (BDE) module.

Actual date

In particular for the goods receipt, it is important to specify the actual date in order to identify the difference between the planned delivery date and the actual delivery date. In production, this field is usually empty.

Planned date

In particular for the goods receipt, it is important to specify the planned date in order to identify the difference between the planned delivery date and the actual delivery date. In production, this field is usually empty.

Status - details

Status

The status specifies the inspection requirement status, e.g. whether it is completed, unchecked ("created" status) or whether inspection results already exist ("partial result" status).

Overall result

The overall result defines whether the inspection requirement is classified as "pass" or "fail".

Usage decision

The usage decision classifies the overall result in more detail. You can extend the list of usage decisions via customizing the system. When you complete an inspection requirement, the usage decision is predefined for each overall result. A "pass" result leads to the usage decision "release" and a "fail" result leads to the usage decision "reject". You can change the predefined usage decision if you manually complete the inspection requirement.

Skip lot

Due to dynamic modifications, an inspection requirement can be classified as "skip lot" and, as a result, no inspection is performed.

PPS/ERP reference

If higher-level systems automatically generate inspection requirements via interface, you can use this reference number to upload and uniquely assign results.

PPS/ERP addition

If higher-level systems automatically generate inspection requirements via interface, you can use this unique reference number to upload results and add an additional information. Only the interface populates this field.

Uploaded to ERP/PPS

If this field is enabled (by interface) it can specify that the upload to the higher-level system has been performed and the higher-level system has confirmed this via interface. Only the interface populates this field.

Inspection plan number/inspection plan index (used)

The inspection plan number and inspection plan index specify the inspection plan that is used as basis for the generation of the inspection requirement and inferior data (inspection steps, inspection step characteristics).

Cavity assignment

You require the authorization "ipl_cav" to display these fields.

The options "none" or "sample" are available for cavity assignment. The respective setting is based on the inspection plan.

Dynamic modification

Dynamic modification type

Shows the dynamic modification type of the respective inspection requirement.

Batch-related: The inspection plan used includes a dynamic modification based on batches.

Characteristic-related: The inspection plan used includes a dynamic modification based on characteristics. In this case, the transitional definition, the initial inspection severity and the dynamic modification norm are specified on the level of inspection step characteristics.

None: The inspection plan used does not include any dynamic modification.

Transitional definition

These fields (number and designation) are only available if dynamic modification based on batches or characteristics is defined for the inspection plan used. The transitional definition provides the possible inspection severities and controls switching between these different inspection severities. The content cannot be changed.

Determined inspection severity

This field defines the inspection severity that is/was used for the inspection of the incoming goods according to the basics for the dynamic modification history. This field is only available if dynamic modification based on batches is defined in the inspection plan used. The content cannot be changed.

Revised inspection severity

This field defines the inspection severity that is/was used for the inspection of the incoming goods according to a manual revision of the inspection severity. This field is only available if dynamic modification based on batches is defined in the inspection plan used.

Form

Party in charge type

Provides the shortlist of the different categories of responsible parties, e.g. department, customer, supplier, external persons.

Party in charge

List of the responsible parties that includes only the previously selected type. This list includes all master data that is marked accordingly in the “party in charge” field.

Party in charge name

Only shows the name/designation of the selected responsible party.

Form

Form type of the inspection plan used.

Calibration

Resource type

Shows the resource type of the resource to be calibrated. For gages, this is the type “PRM”.

Resource

Shows the resource number (gage number) that is assigned to this inspection requirement.

Maintenance

Shows the activity number that is assigned automatically by creating an activity in the activity calendar. This unique ID number identifies or references the respective activity.

Calibration findings

Shows the result entered or identified when the calibration process was completed, e.g. capable without restrictions.

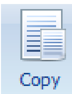
Calibration status

Shows the status entered or identified when the calibration process was completed, e.g. released.

Inspection requirement - editing functions

The key fields "area" and "inspection requirement number" as well as the fields "skip lot" and "uploaded to ERP/PPS" cannot be changed in the editing mode.

Inspection requirement - toolbar



Copy

Function authorization: irp.copy (available as of SP13)

The selected data record is opened in copy mode. To create a new inspection requirement, you must normally change the order number. The data structures that are below the inspection requirement are not copied (inspection steps, characteristics, inspection points, etc.) Upon saving, the system searches for an inspection plan using the specifications of the inspection requirement and creates the respective inspection steps, etc.

Release

Function authorization: irp.release

The system sets the status for the inspection requirement that was available when it was completed, e.g. to the "partial result" status. This is required if the inspection requirement has already been completed and if it must be reset into a status that allows inspection.

Complete

Function authorization: irp.complete

Opens a dialog to assign a usage decision. The system has already preset a usage decision using the available inspection results. The inspection requirement is finally transferred to the "completed" status, once it has been completed. With this status, you can no longer inspect the included inspection steps.

Cancel

Function authorization: irp.cancel

Transfers the inspection requirement to the "canceled" status. With this status, you can no longer inspect the included inspection steps.

Order/operation

Function authorization: irp_goto

Click the respective button to jump to the application.

"Print" detail application

Function authorization	irp.print

The print dialog of the inspection requirement header opens a list of available reports. These are Word forms. The web services that are available in the respective context specify the potential content of these forms. The form entries, i.e. the content of the list of forms of the respective print dialog, are defined within the master data of quality management. Here, the basics for new forms and the respective form properties are also specified. You require the respective license to be able to change the forms with respect to content and design.

The report, 4th edition, 2nd volume VDA (German Association of the Automotive Industry), is available as special form for initial sampling. In addition to the cover sheet, the relevant attachments of the inspection characteristic results are printed as well. Each attachment includes the inspection characteristics and the relevant results for an attached category, e.g. data for dimensional checks (attached category: dimension).

Print – Toolbar

There are no other special function buttons in addition to the standard functions/features.

"Complete inspection requirement“ detail application

Function authorization	irp.complete

To complete an inspection requirement, click the respective button in the toolbar. An editing dialog opens.

The system displays information on the inspection requirement (i.e. order, batch, article, company data) and also information on the inspection steps included in the inspection requirement. You cannot complete an inspection requirement if any inspection steps are not completed, i.e. the field “without result” includes a value greater than zero. In this dialog box, you can complete the inspection orders that are still open using the button “complete all inspection orders”. This is only possible if the conditions of a mandatory inspection that might exist are fulfilled. You can then specify the result and further use of the inspection requirement. If all inspection steps have been completed with “pass”, the system proposes “pass” as the result and “release” as usage. If one inspection step has been completed with “fail”, the system proposes "fail” as the result and “reject” as usage. Another result and usage may still be selected though.

The list specifying the available usage decisions can be changed as part of an MPDV customizing service.

"Inspection step" detail application

In the master detail grid, if you change from an inspection requirement to the inferior level of "inspection steps", the grid displays all inspection steps that are included in the selected inspection requirement. The system displays the data of the selected inspection step in the detail view. The number or the combination of the inspection steps depends on the configurations made for the existing inspection plan. If an inspection plan includes a configuration that specifies that the operation is defined on the level of inspection plan characteristics and if the system configuration specifies in addition that all included inspection steps are created when the inspection requirement and the first inspection step are created, then the inspection requirement includes an inspection step for each operation included in the inspection plan.

You can also release, complete or cancel an inspection step using the toolbar functions.

You can delete inspection steps, but you cannot create or edit them manually.

You can find the current status of the inspection step in the "rating" category of the "inspection step" tab. The following statuses are available:

Status	Description
No characteristic (KMM)	The inspection step has no characteristics that can be inspected. The inspection step is faulty and no inspection can be performed.
Created (ERS)	The inspection step has been created but not yet released for inspection. No measured values have been recorded yet.
Inspection step released (FRE)	The inspection step has been released for inspection. No measured values have been recorded yet.
Partial result (TER)	The inspection step has been released; some measured values have been recorded.
Inspection in progress (IPR)	Measured values for this inspection step are currently being recorded on another terminal.
Completed (ABG)	The inspection step has been completed.
Canceled (STO)	The inspection step has been canceled.
Skip lot (SKL)	The inspection step is in the "skip-lot" status.

"Inspection step" field descriptions

In the following, the most important fields of the *Inspection step* are described. Self-explanatory fields are not explained.

Inspection step

Inspection step number

Unique number of the inspection step

Status

Status description of the inspection step, e.g. "completed", "partial result", etc.

Result

Qualitative classification of the inspection result based on the respective inspection results of the characteristics pertaining to it.

Workplace

Number of the workplace where this inspection step is performed.

Workplace designation

Name of the workplace where this inspection step is performed.

Tab *Inspection point identification*

You can activate up to nine different fields, which can or must be filled out later when inspection points are generated. By customizing the CAQ system options, it may be defined which fields are to be enabled, how they are labeled (for some fields) and whether or not it is a mandatory field.

If a field is activated and it is a mandatory field, the respective checkbox is always "checked". The respective checkbox is checked and grayed out if it is not a mandatory field but enabled for the resulting data collection. If the field is disabled, the checkbox of the field pertaining to it is not activated at all.

Once activated within the inspection point, the fields 1 to 3 are available for entering values with respect to "equipment", "functional location" and "physical sample". The fields 4 to 9 are user fields relating to inspection points. The fields 4 and 5 are "alphanumeric" fields, the fields 6 and 7 are numeric, the field 8 is a "date" field and field 9 is a "time" field.

Order data from inspection step

Order

This is e.g. the production order number in production.

Operation

Number of the operation this inspection step has been generated for. The operation number is derived from the inspection plan used.

Operation name

The operation number, which is derived from the inspection plan used, identifies the operation name. Normally, this field remains empty, because usually only the operation number is used in inspection planning. The operation designation is only used in exceptional cases.

Quantities

Rework

A rework quantity may be entered in this field when the inspection step is completed. This quantity specification is used for information purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. This field is also useful in goods receipt and goods issue.

Scrap

A scrap quantity may be entered in this field when the inspection step is completed. This quantity specification is used for information purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. This field is also useful in goods receipt and goods issue.

Actually produced quantity (act. prod. qty.)

The actually produced quantity may be entered in this field when the inspection step is completed. This quantity specification is used for information purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. This field is also useful in goods receipt and goods issue.

Inspection step - editing functions

You cannot edit an inspection step. You can only add quantity specifications when you complete inspection steps.

Inspection step - toolbar

Release

Function authorization: irisp.release

Sets the status "released" for an inspection step with status "created" or "completed". Or the system sets the status that existed when the inspection step was completed, e.g. the "partial result" status.

Complete

Function authorization: irisp.complete

Opens a dialog where the (inspection) result is preset. The result is based on the existing inspection results of the characteristics. The inspection step is completed and the status "completed" is set. You can no longer perform inspections for inspection steps with this status.

Cancel

Function authorization: irisp.cancel

Sets the "canceled" status for the inspection step. You can no longer perform inspections for inspection steps with this status.

Order/operation

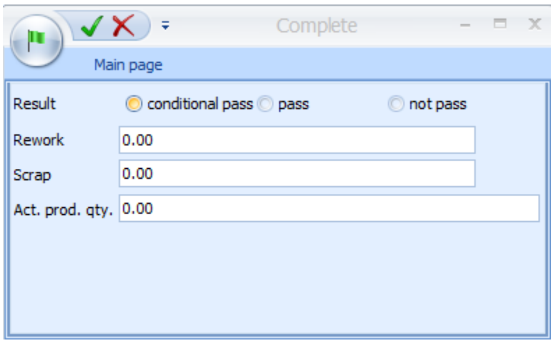
Function authorization: irp_goto

Click the respective button to jump to the application.

"Complete inspection step" detail application

Function authorization	irisp.complete

To complete an inspection step, click the respective button in the toolbar. The following editing dialog opens.



If all inspection step characteristics have the result "pass" or if they are not checked, the system suggests the result "pass". The system suggests the result "fail" if there is at least one inspection step characteristic that has the inspection result "fail". The user can also select another result.

When the inspection step is completed, the user can record a "rework quantity", "scrap quantity" and the "actually produced quantity" (not relevant for gage management). This quantity specification is used for information purposes only and is not processed by default. Normally, this field is only used if the shop floor data collection module (BDE) is not used in production, i.e. only the CAQ module is in use. These fields are also useful in goods receipt and goods issue. In particular in the goods receipt area, it is useful to record scrap and rework. These details can be uploaded for several inspection steps to a higher-level system at a later point in time when inspection requirements are completed.

"Inspection points" detail application

By default, the system only uses inspection points in production. It is only useful to have inspection points if you take and check several samples for one order/inspection requirement. For example, this is useful if you perform inspections in production that are based on different events. Example of an event: A specified inspection interval that is based on quantities or time is reached. The events that trigger inspections can be different for the individual inspection plan characteristics. The inspection points of an inspection step can therefore have different inspection characteristics each.

Once an inspection has been performed, you must complete each inspection point. Each inspection point has an inspection result (pass, fail) and a status (completed, open). Inspection points that have not yet been checked have a "pass" inspection result by default.

Note

If you enter a text in the inspection point fields on the AIP terminal that is longer than the field provided in the respective inspection point dialog, the text is cut off when you save the inspection point on the AIP. Therefore, the field contents should not be longer than the field lengths configured for the AIP inspection point dialog. The field lengths of AIP dialogs can be adjusted as part of system customization.

"Inspection points" field descriptions

In the following, the most important fields of the *Inspection step* are described. Self-explanatory fields are not explained.

Inspection point

Inspection point number

The inspection point number uniquely identifies the inspection point of the inspection step. In addition to the "inspection point" field, the higher-level key fields "inspection step number" and "inspection requirement number" are shown .

Inspection result

The inspection result can be "pass" or "fail". Inspection points that have not yet been checked have the inspection result "pass" by default (standard system configuration).

Status

An inspection point can be completed or open. Only open inspection points can still be checked on the AIP terminal.

Tab *Inspection point identification*

This tab displays the fields, which have been enabled in the higher-level inspection step. You define in the CAQ system options which fields identify the inspection point and which fields are mandatory fields. By default, the following fields are displayed, but none of these fields is mandatory.

Date

You can record a date. If you create a new inspection point, the system date and time is preset. If you manually create the inspection point, you can directly change date and time. Or you can change date and time in the editing mode.

Time

You can record a time. If you create a new inspection point, the system date and time is preset. If you manually create the inspection point, you can directly change date and time. Or you can change date and time in the editing mode.

In addition to the configurable identification fields, there are still the following identification fields that are always available.

Workplace

If the inspection point is generated on an AIP terminal or if the inspection point is automatically generated via inspection event on a specific terminal, this field shows the workplace of the triggering inspection station. If you specify an inspection station, this inspection point is only available at this inspection station.

ERP batch

You can enter a respective ERP batch.

Partial batch

You can enter a respective partial batch.

Details

Generation of inspection points

If the system automatically generates an inspection point because a defined time or quantity interval is reached, the inspection point has the identifier "relating to time" or "relating to quantities". An inspection point that is assigned the identifier "free" has been generated manually or on the basis of other events that make inspections become due.

Inspection points - editing functions

You cannot create new inspection points for completed or canceled inspection steps. And you cannot edit existing inspection points if the respective inspection step is completed of canceled. If you want to create or edit an inspection point, you must first release the inspection step.

Inspection points - toolbar

Release

Function authorization: iripp.release

Sets the "open" status for a completed inspection point.

Complete

Function authorization: iripp.complete

Opens a dialog window where you can complete an inspection point.

Order/operation

Function authorization: irp_goto

Click the respective button to jump to the application.

"Complete inspection point" detail application

Function authorization	iripp.complete

To complete an inspection point, click the respective button in the toolbar. The following editing dialog opens. The dialog includes an inspection result that has been calculated using the results of the inspected characteristics. This result cannot be changed. The category "Rating" also includes a preset usage decision for the inspection point that is based on the inspection result of the characteristic. You can manually change the usage decision. The system provides a selection list for usage decisions of inspection points that is filtered by the preset production site. Each usage decision classifies the inspection point as "pass" or "fail", irrespective of the inspection result. When you have selected a usage decision, the system displays additional information on the usage decision of the inspection point. This additional information is, for example, the code group, code number, catalog type and the factory/site.

MPDV services for customizing the system may provide further usage decisions for inspection points.

If the identification fields are configured accordingly, you can edit these fields when you complete the inspection point. If an identification field is a mandatory field, you must enter a value before you can complete the inspection point. In addition to this, quantity fields may also be filled out.

"Inspection step/inspection point characteristics" detail application

Function authorization	iriscp.*

The detail application of inspection steps/inspection point characteristics is almost identical to the master data application of characteristics. Therefore, only additional features or modifications are described here.

Go to

For further information on the definition of characteristics, refer to the functional description in the document MOC_CharacteristicsQM.

The master detail grid shows the inspection step characteristics one level below the inspection step. If you use inspection points and if you have created inspection points, you can open the list of inspection points one level below the inspection steps. If you open this list, the characteristics of the selected inspection point are displayed in a separate tab.

The characteristics of the inspection step are specified in the inspection plan used. If the inspection plan settings specify that a separate inspection step must be generated for each operation (global setting for all workplaces), then the inspection step only includes the characteristics of a specific operation. If the inspection plan settings additionally specify that a separate inspection step must be generated for each workstation, then an inspection step always includes the characteristics of a fixed combination of operation and workplace.

An inspection point only includes characteristics that become due because of the same event. Also the manual generation of inspection points represents a due date event. For example, if a quantity interval defined for the inspection step characteristics has been reached, an inspection point is automatically generated. This inspection point includes all characteristics that have the same quantity interval. And it also includes the characteristics that have a quantity interval that is the least common multiple of the quantity interval reached.

Example:
Characteristic 1 has a quantity interval of 100. Characteristic 2 has a quantity interval of 200. After the first 100 parts produced, an inspection point is generated, which only includes characteristic 1. After 100 more parts, another inspection point is generated, which includes the characteristics 1 and 2.

If an inspection becomes due because the machine status changes, the generated inspection point only includes characteristics that must be checked because of the same machine status change.

Only manually generated inspection points are not limited with respect to the respective characteristics. These inspection points always include all characteristics of the related inspection step.

Inspection step/inspection point characteristics - field descriptions

In the following, only the fields are described that are not included in the master data of characteristics or in the inspection plan characteristics.

Characteristic

Inspection result (available as of SP13)

The inspection result is identified using the settings for the characteristics in the inspection plan according to the "Inspection result base" and the value of the CAQ option 1038. For details, refer to the description of the CAQ option 1038 in the procedure document "Configuration_QM_Options".

Initial sample creation

Shows the field content that is transferred from the characteristic included in the inspection plan used. A separate sheet/page is printed for each category in the printed initial sample report.

Properties

Dynamically modified

This checkbox is enabled, provided that a characteristic is dynamically modified.

Dynamic modification

Determined inspection severity/inspection severity designation

This field shows the inspection severity identified for this characteristic on the basis of the dynamic modification rules specified in the inspection plan.

Revised inspection severity/inspection severity designation

This field only exists if dynamic modification is based on characteristics. A revised inspection severity is displayed here, if the user has changed the process of dynamic modification and if the user has revised the current inspection severity (new initial inspection severity).

Note

A note is displayed in this field, in case an error has occurred with the dynamic modification of this characteristic while inspection steps are generated.

Details

No cavity

You require the authorization "ipl_cav" to display these fields.

You can use this option to label a characteristic as not relevant to cavities, although the respective inspection requirement specifies that the relevant characteristics are (actually) to be recorded in relation to cavities.

You must always enable this checkbox for attributive characteristics and inspection charts.

Sample group

This field specifies which sample group this characteristic belongs to.

Sampling

If a characteristic has been assigned to a sample group, this option specifies if you use this characteristic to generate a sample. In this case, you call it a sampling characteristic. You must enable the checkbox in this case. You must assign a sample group to the sampling characteristic.

If these fields are not available, you require a new program version of this application.

Specifications

Sample size

This field shows the identified sample size if a sampling scheme is defined in the inspection plan characteristic and if this sampling scheme does not specify the sample size because the sample size is calculated using the actual quantity of the inspection requirement. This is the case for the sampling schemes "batch inspection" or "100% inspection".

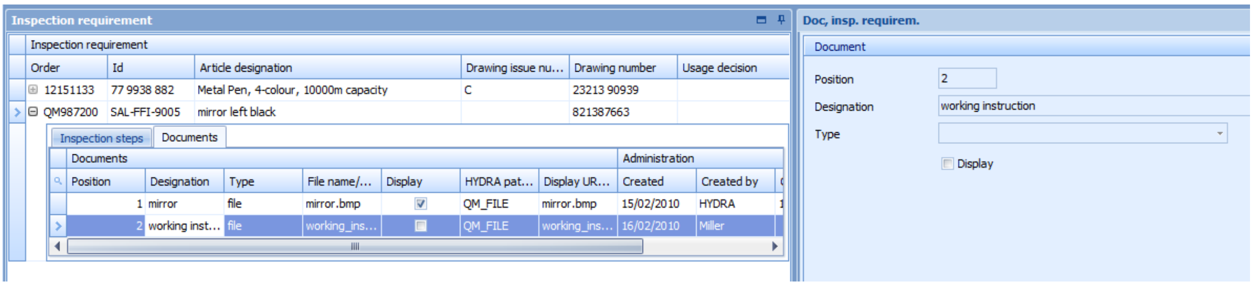
Go to

The other fields included in the tabs are the same as the fields of the characteristic master data and are described in the documentation "[CAQ characteristic master data]".

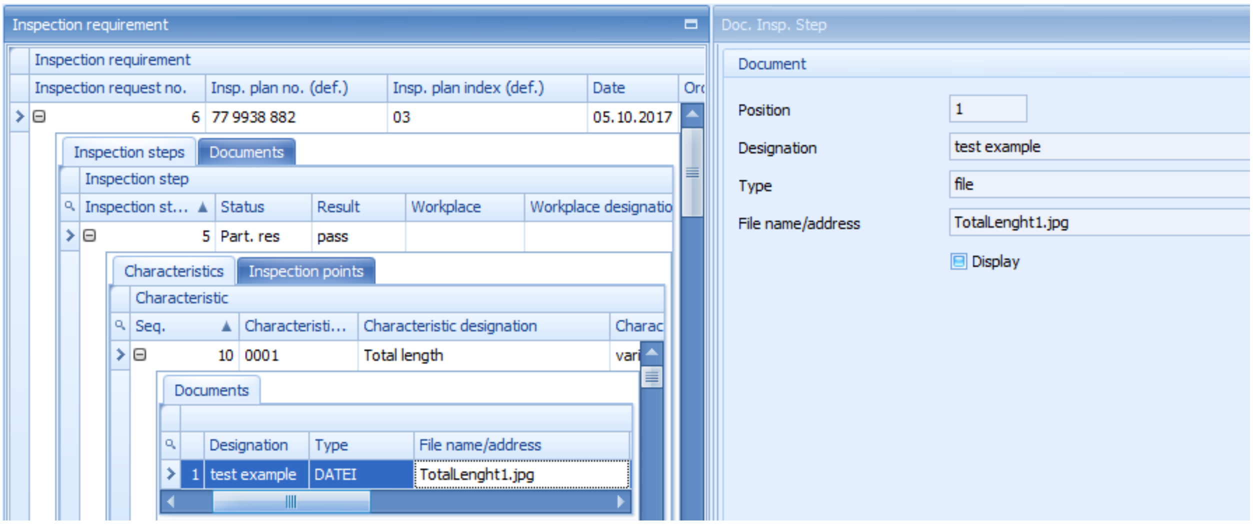
Inspection step/inspection point characteristics - editing functions

Inspection step or inspection point characteristics cannot be edited.

"Inspection requirement documents" and "Characteristic documents" detail applications



The above screenshot shows how an inspection plan document is assigned.



The above screenshot shows how a characteristic document is assigned.

If you have enabled the "documents" tab in the master-detail grid, you can assign any number of documents to each inspection requirement and to each inspection step characteristic. If this tab is enabled in the master-detail grid, the respective editing buttons are enabled in the toolbar to edit the documents.

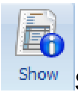
You can use all formats registered by Windows for the documents you want to assign. You can therefore assign simple documents (e.g. written in Word), drawings of any format and videos. You only have to make sure to install a program that is able to display the used format. The appropriate program linked in Windows opens the documents.

The file types "File", "URL", and "Text" are available. For the type "file", you can manually enter the file name including the path. The file type “URL” enables access to the internet or intranet. With the file type "text", you can directly enter a text.

You can assign a designation/name to each document that is added. You can also define the list order of the documents. Use the field "position" to define the order (numeric input). Position numbers must be unique in this list. Enable the checkbox *Display* to define that the document is displayed during inspection process.

"Inspection plan documents“ and "Documents for inspection plan characteristics" – Toolbar

In addition to the standard functions, the button to show documents is available.



Show documents

If a document link is stored, this button opens and shows the linked document. However, a program, which can show the linked file type, must be installed in the PC.

"Failure" detail application

A "Failure" detail application is provided on the level of inspection requirements. This list shows all failures of the type "failure type", "failure location", "failure cause" and "originator", which have been assigned to the characteristics, samples or measured values during the inspection process. The list even shows the failure types that have been generated automatically, e.g. "upper tolerance limit violated".

In addition to the respective failure, the following referenced data is also available.

- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number
- Failure type
- Characteristic number and description
- Workplace number and description
- Weighting (number, e.g. for an inspection chart characteristic)

- Comment,
- Failure date
- Failure time

You can restrict the list in a flexible way using the individual column filter. You can also use the *group by* function. Example: You can list the failures for each characteristic.

Failure list - editing functions

For administration purposes, you can create, edit or delete failure analysis entries for inspection requirements. If a new data record is created, you can enter further key information to specify the assignment. The following information entries can be added:

- Inspection step number,
- OP sequence number (uniquely identifies the characteristic),
- the sample number
- and value number.

To precisely assign this key information, the user requires "administrative" skills. An administrative environment is recommended.

The user requires the following function authorizations:

Function authorization	failure.create => create new failure failure.edit => edit failure failure.delete => delete failure

Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and a validation process is not performed.

The fields "area" and "inspection requirement number" cannot be modified and you cannot enter values. This is also true when you create new data records. When you create a new data record, this data is taken from the previously selected inspection requirement.

You can enter values for all other fields when you create a data record. In the editing mode, you can only change the fields "weighting" and "comment".

"Measures" detail application

A detail application including a "measure list" is provided on the level of inspection requirements. This list shows all measures and actions which have been assigned to the characteristics, samples or measured values during the inspection process.

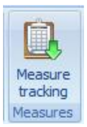
In addition to the respective measure number and designation, the following data is also referenced.

- Inspection requirement
- Inspection step
- Operation sequence number (OP sequence, AFO)
- Sample number
- Value number
- Measure type
- Party in charge incl. type
- Status
- Comment
- Text
- Effectiveness

You can restrict the list in a flexible way using the individual column filter. Use the "group by" function to have further options of analysis.

"Measures" toolbar

Use the button "measure tracking" to open the measure tracking dialog where you can edit and create measures. Measure tracking uses the internal and unique "serial number" of the measure in the "references" tab for filtering. Filtering is omitted if no measure is selected.



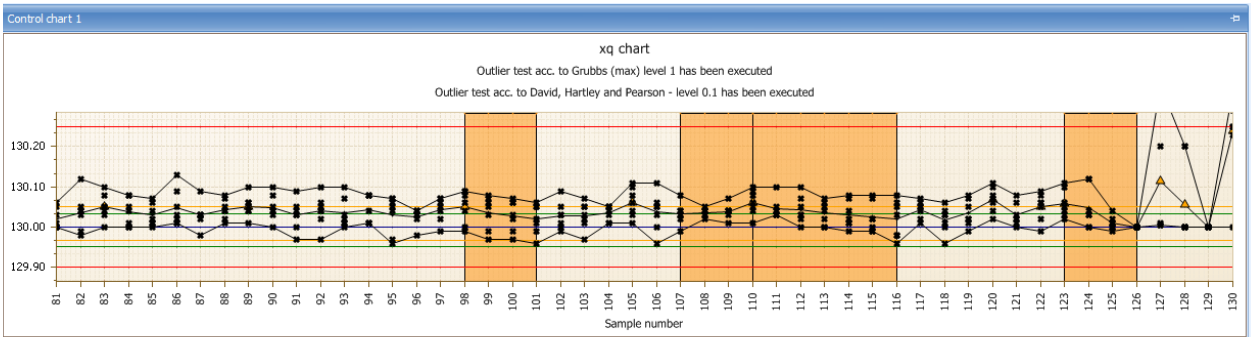
Go to measure tracking

Function authorization	failure.* => opens measure tracking

"Control chart 1/2" detail application

By default, the application shows the control charts 1 and 2 defined for the inspection step characteristic. If the charts are not defined for the inspection step characteristic, the xq and s charts are used for variable characteristics and the p and u chart for attributive characteristics.

All data of the selected inspection step characteristic is used for the graphic presentation.



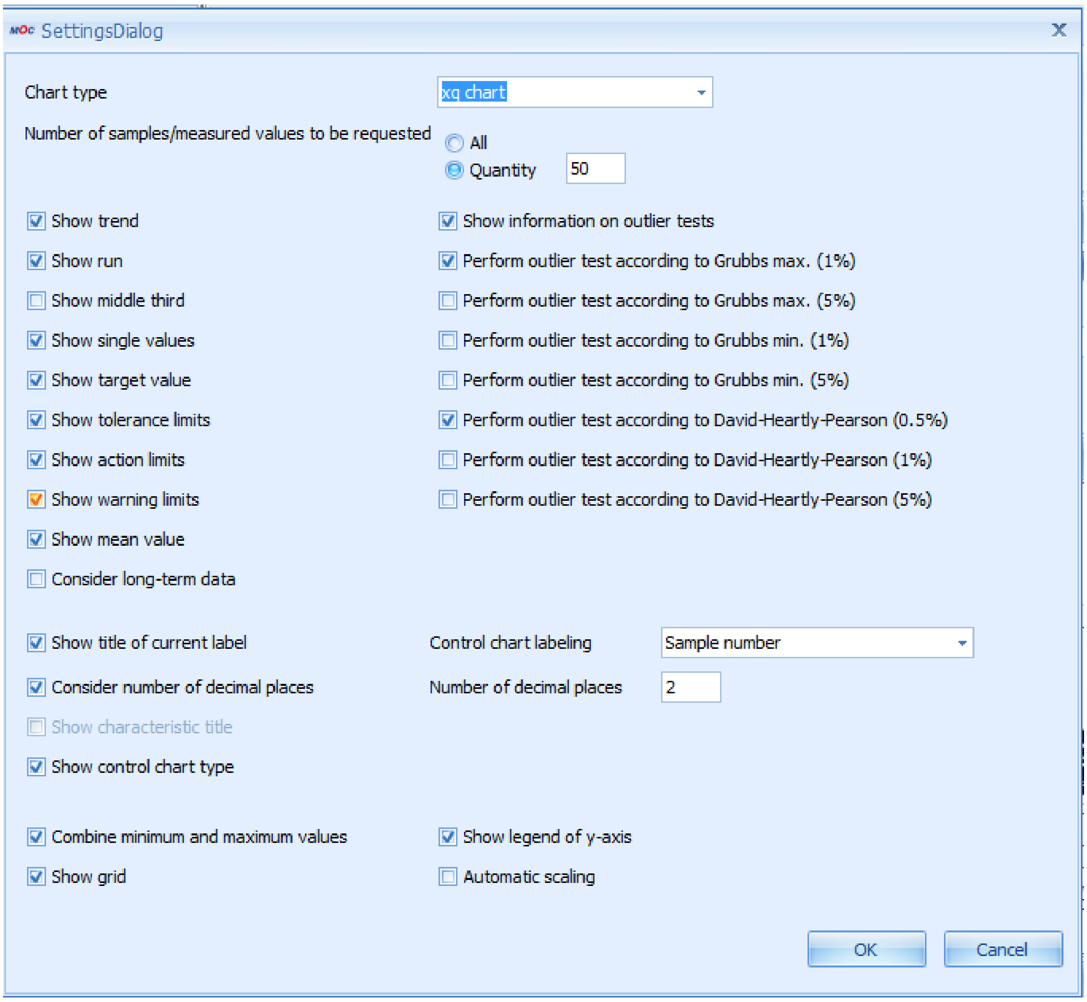
Open the dialog to configure "control chart 1" or "control chart 2" to define the contents of this application. If you use this dialog to make changes, the changes are saved per user. If no specific setting has been made, the xq chart ("variable" characteristic type) or the p chart ("inspection chart" or "attributive" characteristic type) is displayed by default. The available control charts depend on the characteristic type. By default, the user can select one of the following control charts.

Variable characteristic

- Xq chart
- S chart
- R chart
- Single value chart
- Median chart

Attributive characteristic

- p chart
- np chart
- c chart
- u chart



In the follolwing, the most important configuration options are described:

Number of samples/measured values to be requested

Specifies the number of samples or single measured values that are displayed in the control chart.

Show …

The options that include the term "Show ..." enable or disable the display of the respective data in the control chart.

Consider long-term data

Check this option to integrate archived data of the medium-term data area.

Combine minimum and maximum values

It is recommendable to show the minimum and maximum values that are each connected by a line to improve the presentation of the range of dispersion of single values.

Automatic scaling

Use the "automatic scaling" function to display all values, irrespective of the existing limit values. This function also shows extreme outliers in the control chart. Disadvantage: The other measured values are much smaller in the layout and it is more difficult to identify changing values or a trend.

Show trend / run / middle third

Use the monitoring functions "trend", "run" and "middle third" to better monitor a process. These functions are only available with the control charts xq and median. If you show the trend, you can visualize a process trend that may rise or fall. The system uses several samples to generate a trend. By default, these are seven consecutive rising or falling values. The run shows sections where the process runs above or below the mean value (when displayed, otherwise the target value). The run covers several samples. By default, these are seven consecutive values that are above the mean value. The default number of seven samples/values that is set to identify a trend or run can be changed via system customization. The system identifies a "middle third", if an unusually high or low number of values lies within the middle third of the range between the action limits.

The system automatically makes the respective analyses for "trend", "run" and "middle third". If a trend/run and/or middle third is identified, the control chart shows it in a graphic form. The following events are shown in the control chart via symbols or colors:

- Trend (colored area)
- Run (colored area))
- Middle third (colored area)
- Outlier (symbol: )
- Xq violates action limit (symbol )

The presentation of outliers is only possible in the xq chart and median chart. Outliers can only be displayed when single values are shown. The function to perform outlier tests and the function to display outliers are separated. You can activate the different outlier tests for the different levels separately. If you have enabled the display of outlier tests, the test result is shown in text form on top of the respective control chart.

The below outlier tests are available for the different inspection levels. The inspection level is specified in parentheses.

- Grubbs max. (1 %)
- Grubbs max. (5 %)
- Grubbs min. (1 %)
- Grubbs min. (5 %)
- David-Hartley-Pearson (0.5 %)
- David-Hartley-Pearson (1 %)
- David-Hartley-Pearson (5 %)

Notes on outlier tests

- The outlier tests do not refer to the total of all samples, i.e. each sample is considered individually. Consequently, the following phenomena may appear:
    - Despite a large range between minimum and maximum value no outlier can be identified. A reason for this may be the equal distribution of the single values within the sample.
    - Despite a low range between minimum and maximum value, an outlier is identified. A reason for this may be an accumulation of many values at one "point" so that an individual value having a certain distance to this agglomeration is identified as outlier within the sample.
- To perform an outlier test, there must be at least three values in the sample. The larger the number of values in a sample, the more uniform is the overall picture of the outliers compared to all samples.
- The Grubbs outlier test is performed with a sample size of $2 < n < 148$ (n = number of values within a sample). The outlier test according to David, Hartley und Pearson is performed with a sample size of $2 < n < 1251$ (n = number of values within a sample).

X-axis labeling

The below information is available to label the x-axis of control charts.

- Sample number
- Order number
- PPS reference number
- Purchase order number
- ERP batch
- Article number
- Machine number
- Cavity number
- Date + time of the first measured value of a sample
- Date + time of the last measured value of the sample
- Date + time of sample completion
- Badge number of the first measured value of the sample
- Badge number of the last measured value of the sample
- Badge number of sample completion

If the following fields are not available, you require a new program version of this application:

- Partial batch
- Workplace

- Production workplace
- Field 1
- Field 2
- Field 3
- Field 4
- Field 5
- Field 6
- Field 7
- Field 8

The fields "field 1" to "field 8" are enabled as part of an MPDV customizing. The field labels change according to the individual requirements. Because these field names are flexible, they are only labeled "field 1" to "field 8". Field 1 includes, for example, the tool if cavity-related data collection is enabled.

Tooltips in the control chart

If the system has identified a value as outlier, the detailed information is shown when you mouse over the symbol (red rhomb). The tooltip includes the test(s) that issued this outlier result.

For mean values, the tool tip shows the value and date and time of the first and last measured value of this sample and the respective inspection requirement number and inspection step number.

For single measured values, the tooltip shows the exact value.

For the colored areas that identify a trend, run or middle third, a respective note including the reason is shown .

Sorting of measured values

The server decides on the sorting of a control chart. The sorting type depends on the settings of control chart filters. If the filters "operation sequence" and "inspection requirement number" or "operation sequence" and "inspection step number" are entered, the characteristic is identified uniquely and sorting is based on the sample number.

Sorting is based on date and time if either the filter field "operation sequence" is left empty or if it is filled and the fields "inspection requirement number" and "inspection step number" are left empty.

The displayed sorting of measured values influences the visual presentation of a trend or run. However, the automatic generation of the failure type "trend" is always based on the measured values being sorted by the sample number, as the numbers of the operation sequence, the inspection requirement and the inspection step are always known at the time data is recorded.

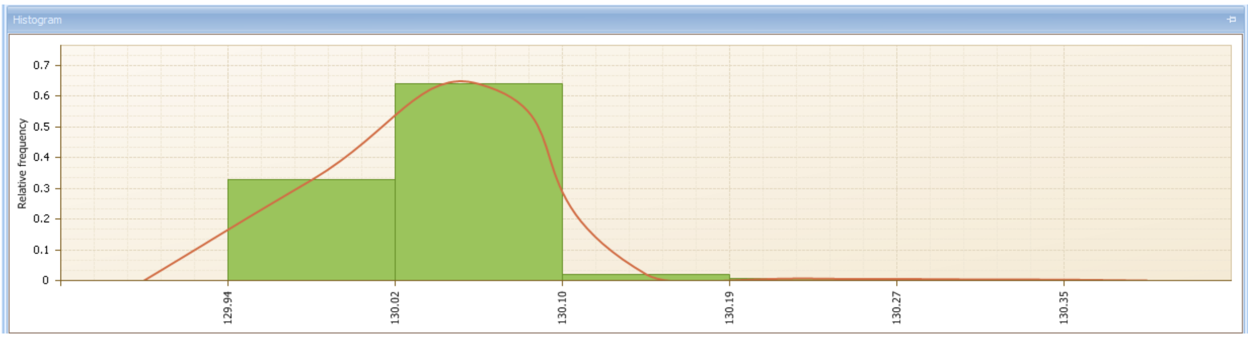
Consequently, the following scenario might be possible.

- The control chart is sorted by date and time and a trend is shown, although the automatic failure "trend" has not been generated.

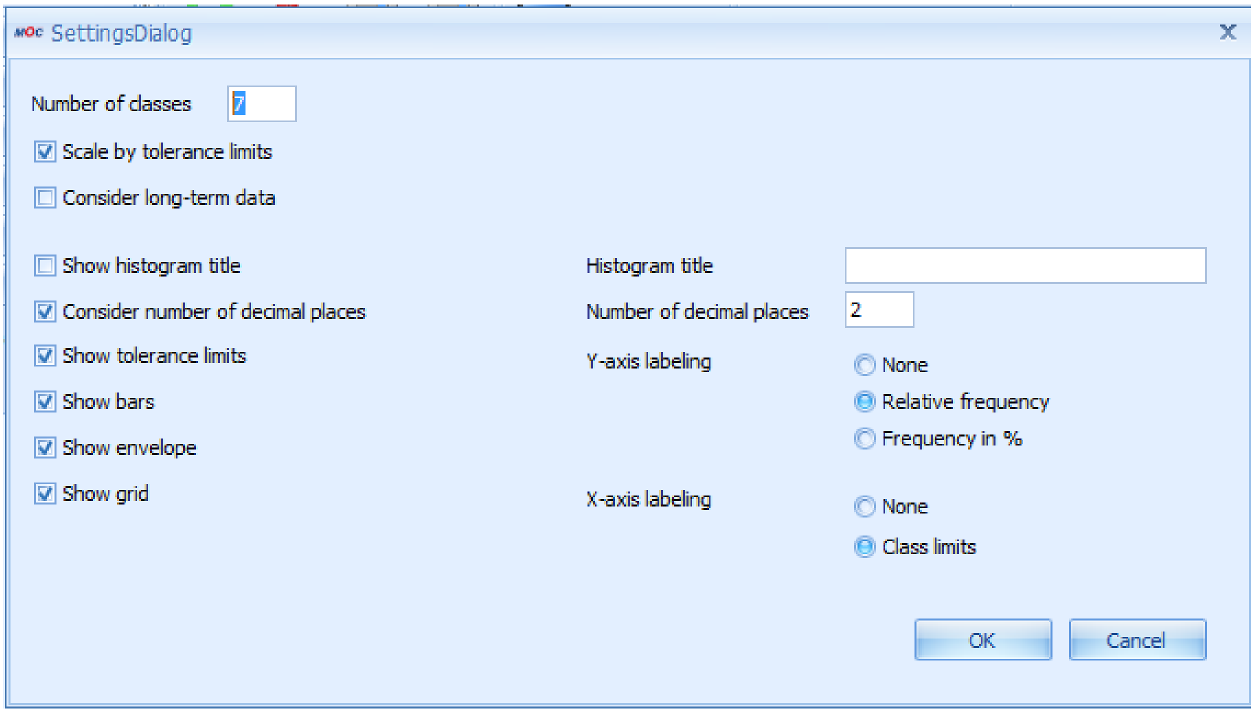
The control chart is sorted by the sample number and no trend is shown, although the automatic failure "trend" has been generated.

"Histogram" detail application

All data of the selected inspection step characteristic is used for the graphic presentation.



You can restrict the number of samples displayed in a control chart. This is not possible with a histogram. The histogram is always based on the total of available samples matching the selection filter criteria. The number of classes and the additionally displayed information influence the histogram presentation. The contents of this application are defined by opening the dialog to configure the "histogram". If you use this dialog to make changes, the changes are saved per user.



In the follolwing, the most important configuration options are described:

Number of classes

Specifies the number of histogram classes. The measured values are classified and displayed in these classes. If the histogram shows the values within the tolerance limits (option "Scale by tolerance limits"), two histogram classes include the values exceeding the tolerance limit (upper and lower).

Scale by tolerance limits

Enabled: The classes are between the tolerance limits - two "outlier classes" being displayed, one to the left and one to the right.

Disabled: The classes include the total range of all measured values (no separate classes for values exceeding the tolerance limits).

Consider long-term data

Includes the archived data of the medium-term data area.

Show histogram title

Enable this option if you want to show a special histogram title.

X-axis labeling

If the option "Class limits" is set, the x-axis shows the respective values of the class limit. You can use the two configuration options for decimal places to specify the detail, i.e. the number of decimal places.

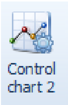
Consider the number of decimal places

With this option, the defined number of decimal places is used for the x-axis labeling.

Control chart 1/2 and histogram - toolbar

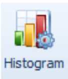
Control chart 1 settings

Opens a dialog to configure the settings of control chart 1. The respective details are described in the respective detail application.



Control chart 2 settings

Opens a dialog to configure the settings of control chart 2. The respective details are described in the respective detail application.



Histogram settings

Opens a dialog to configure histogram settings. The details are described in the respective detail application.

"Samples" detail application

The "samples" detail application is on the level of inspection step characteristics. It includes statistical key figures for each sample. In addition to the referenced key fields, such as the inspection requirement number, inspection step number and sample number, the following statistical values are listed if available/calculated:

- Xq
- Xq floating
- R floating
- Standard deviation s
- s floating
- Minimum
- Maximum
- Range
- Median
- P

- U
- Number of defects

The list additionally shows the characteristic specifications (tolerance, action and warning limits) as well as the referenced machine.

"Samples" - toolbar

Complete sample

It might be necessary to complete samples manually, because measured values can be entered for administration purposes or because there might be inspections without inspection points but including the initial creation of samples and/or inspections without having reached the sample size. You can also complete samples that have not been completed by the AIP inspection data collection.

Release sample

You can use this function to re-release samples and to record further measured values for this sample number if the specified sample size has not yet been reached.

"Single value" detail application

The "Single value" detail application is on the level of the inspection step characteristics and includes single measured values for all variable inspection step characteristics. For attributive characteristics the following information is displayed:

- Number of defects
- Number of NCU (non-conforming units) and
- Failure

You can also identify if the measured value or the attributive assessment is valid or invalid.

In addition to the referenced key fields such as inspection requirement number, inspection step number, sample number and value number, the detail application lists the characteristic specifications (tolerance, action and warning limits). For each entry, you also have details on the date and time when the entry was recorded and edited and on the responsible user.

Single value - editing functions

For administration purposes, you can create, edit or delete measured values for a variable inspection step characteristic and/or inspection point characteristic. Or you can create, edit or delete the number of non-conforming units for attributive characteristics. If a new data record is created, you can enter further key information to specify the assignment. The following information entries can be added:

- the sample number
- and value number.

To precisely assign this key information, the user requires "administrative" skills. An administrative environment is recommended. Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and a validation process is not performed.

The user requires the following function authorizations:

Function authorization	value.create => create new measured value value.edit => edit measured value value.delete => delete measured value

Users must have profound knowledge of inspection processes, as they are not guided through the process of data collection/modification and a validation process is not performed.

The fields

- Area,
- Inspection requirement number,
- Inspection step no.,
- Inspection step number (only available for inspection point characteristics),
- OP sequence (uniquely identifies the characteristic),
- Upper tolerance limit,
- Target value,
- Lower tolerance limit and
- Number of decimal places

cannot be modified and you cannot enter values. This is also true when you create new data records. When you create a new data record, this data is taken from the previously selected characteristic and the referenced inspection requirement as well as from the inspection point.

You can enter values for all other fields when you create or edit a data record.

The field "single value of cavity" is only visible if

- the inspection requirement includes a setting specifying that data is generally collected in relation to cavities and
- the relevant characteristic does not include a setting specifying that the inspection must be performed without relation to cavities.

If a new data record is created, the field for the measured value will be restricted to the number of decimal places. The other decimal places are filled up with zeros. For techical reasons, these "additional" decimal places are shown during processing.

The field "measured value" is shown for variable characteristics. Instead of the measured value field, the fields "sample size (checked)", and "number of DU" are shown for attributive characteristics, i.e. this also includes inspection chart characteristics.

Detail application "Statistics"

The detail application "Statistics" is on the level of inspection step characteristics. It includes the usual statistical key figures for the selected characteristic.

For variable characteristics, the following statistical key figures are included:

- Number of samples
- Number of measured values
- Xqq
- Minimum
- Maximum
- R
- S
- S relative
- Sigma
- Cp and
- Cpk

For attributive characteristics, the following statistical key figures are included:

- Number of non-conforming units
- Number of defects
- p and
- u.